

```
##Midha Assignment Week 3
```

```
## 1A:
```

```
attach(iris)
```

```
hist(Sepal.Width)
```

```
## 1B
```

```
#based on the plot above. the data seems to be pretty symmetric, if anything slightly skewed to the right, with the mean slightly higher than the median.
```

```
## 1C
```

```
mean(Sepal.Width)
```

```
median(Sepal.Width)
```

```
#looking at the above information, my initial expectation of the data being skewed slightly right was correct with the mean 0.06 higher than the median.
```

```
## 1D
```

```
#finding the width where only 23% of the data is higher than is the same as asking for the 73 percentile, found below.
```

```
quarts=quantile(Sepal.Width,.73)
```

```
## 1E
```

```
par(mfrow = c(2,3))
```

```
plot(Sepal.Width~Sepal.Length, xlab='Sepal Length', ylab = 'Sepal.Width', main='Sepal Width Vs Sepal Length')
```

```
plot(Sepal.Width~Petal.Width, xlab='Petal.Width', ylab = 'Sepal.Width', main='Sepal Width Vs Petal.Width')
```

```
plot(Sepal.Width~Petal.Length, xlab='Petal.Length', ylab = 'Sepal.Width', main='Sepal Width Vs Petal.Length')
```

```
plot(Sepal.Length~Petal.Width, xlab='Petal.Width', ylab = 'Sepal Length', main='Sepal Length Vs Petal.Width')
```

```
plot(Sepal.Length~Petal.Length, xlab='Petal.Length', ylab = 'Sepal Length', main='Sepal Length Vs Petal.Length')
```

```
plot(Petal.Width~Petal.Length, xlab='Petal.Length', ylab = 'Petal.Width', main='Petal.Width Vs Petal.Length')
```

```
par(mfrow = c(1,1))
```

```
## 1F
```

```
#the two plots with the strongest relationship appear to be comparing Petal Width and Length, as well as comparing Petal Length to Sepal Length. These plots have a strong line that indicates more of a relationship between the data sets being plotted. The two plots that do not have a strong connection are comparing Sepal Length to Sepal Width as well as comparing Sepal Width to Petal Width. These plots seem to have a more scattered arrangement of data indication there is no strong relationship.
```

```
## 2A
```

```
attach(PlantGrowth)
```

```
hist(weight, breaks = seq(3.3,6.6,.3), xaxt='n', main='Plant Growth Weight')
```

```
axis(1,at = seq(3.3,6.9,.3))
```

```
## 2B
```

```
boxplot(weight~group, col=2:4)
```

```
## 2C
```

```
# looking at the plots, about 75% of the trt1 weights are below the minimum value of the trt2 weights
```

```
## 2D
```

```
trt2 = weight[c(21:30)]
```

```
min_trt2 = min(trt2)
```

```
trt1 = weight[c(11:20)]
```

```
mean(trt1 < min_trt2) * 100
```

```
## 2E
```

```
HeavyWeight = c(0,1)
for (i in 1:length(weight)){
  if (weight[i] > 5.5){
    HeavyWeight[i] = weight[i]
  }
}
HeavyWeight_clean = HeavyWeight[!is.na(HeavyWeight)]
HeavyWeight_clean = HeavyWeight_clean[2:9]
HeavyWeight_clean

barplot(HeavyWeight_clean, xlab='Flowers', ylab='Weight', main='Boxplot of Flowers Heavier
than 5.5', col = hcl.colors(8, "Viridis"))
```

```
> source("~/GRAD505/Assignment-2/Week3_Assignment_Midha.R", echo = TRUE)
```

```
> ##Midha Assignment Week 3
```

```
> ## 1A:
```

```
> attach(iris)
```

The following objects are masked from iris (pos = 4):

Petal.Length, Petal.Width, Sepal.Length, Sepal.Width, Species

The following objects are masked from iris (pos = 6):

Petal.Length, Petal.Width, Sepal.Length, Sepal.Width, Species

The following objects are masked from iris (pos = 8):

Petal.Length, Petal.Width, Sepal.Length, Sepal.Width, Species

The following objects are masked from iris (pos = 11):

Petal.Length, Petal.Width, Sepal.Length, Sepal.Width, Species

```
> hist(Sepal.Width)
```

Hit <Return> to see next plot:

```
> ## 1B
```

```
> #based on the plot above. the data seems to be pretty symmetric, if anything  
slightly skewed to the right, with the mean slightly higher tha .... [TRUNCATED]
```

```
[1] 3.057333
```

```
> median(Sepal.Width)
```

```
[1] 3
```

```
> #looking at the above information, my initla expectation of the data being  
skewed slightly right was correct with the mean 0.06 higher than the medi ....  
[TRUNCATED]
```

```
> ## 1E
```

```
> par(mfrow = c(2,3))
```

```
> plot(Sepal.Width~Sepal.Length, xlab='Sepal Length', ylab='Sepal.Width',  
main='Sepal Width Vs Sepal Length')
```

Hit <Return> to see next plot:

```

> plot(Sepal.Width~Petal.Width, xlab='Petal.Width', ylab ='Sepal.Width',
main='Sepal Width Vs Petal.Width')

> plot(Sepal.Width~Petal.Length, xlab='Petal.Length', ylab ='Sepal.Width',
main='Sepal Width Vs Petal.Length')

> plot(Sepal.Length~Petal.Width, xlab='Petal.Width', ylab ='Sepal Length',
main='Sepal Length Vs Petal.Width')

> plot(Sepal.Length~Petal.Length, xlab='Petal.Length', ylab ='Sepal Length',
main='Sepal Length Vs Petal.Length')

> plot(Petal.Width~Petal.Length, xlab='Petal.Length', ylab ='Petal.Width',
main='Petal.Width Vs Petal.Length')

> par(mfrow = c(1,1))

> ## 1F
> #the two plots with the strongest relationship appear to be comparing Petal
Width and Length, as well as comparing Petal Length to Sepal Len ....
[TRUNCATED]
The following objects are masked from PlantGrowth (pos = 4):

    group, weight

The following objects are masked from PlantGrowth (pos = 6):

    group, weight

The following objects are masked from PlantGrowth (pos = 8):

    group, weight

The following objects are masked from PlantGrowth (pos = 10):

    group, weight

> hist(weight, breaks = seq(3.3,6.6,.3), xaxt='n', main='Plant Growth Weight')
Hit <Return> to see next plot:

> axis(1,at = seq(3.3,6.9,.3))

> ## 2B

```

```
> boxplot(weight~group, col=2:4)
```

Hit <Return> to see next plot:

```
> ## 2C
```

```
> # looking at the plots, about 75% of the trt1 weights are below the minimum  
value of the trt2 weights
```

```
>
```

```
> ## 2D
```

```
> trt2 = weight[c(21:30)]
```

```
> min_trt2 = min(trt2)
```

```
> trt1 = weight[c(11:20)]
```

```
> mean(trt1 < min_trt2) * 100
```

```
[1] 80
```

```
> ## 2E
```

```
>
```

```
>
```

```
> HeavyWeight = c(0,1)
```

```
> for (i in 1:length(weight)){
```

```
+   if (weight[i] > 5.5){
```

```
+     HeavyWeight[i] = weight[i]
```

```
+   }
```

```
+ }
```

```
> HeavyWeight_clean = HeavyWeight[!is.na(HeavyWeight)]
```

```
> HeavyWeight_clean = HeavyWeight_clean[2:9]
```

```
> HeavyWeight_clean
```

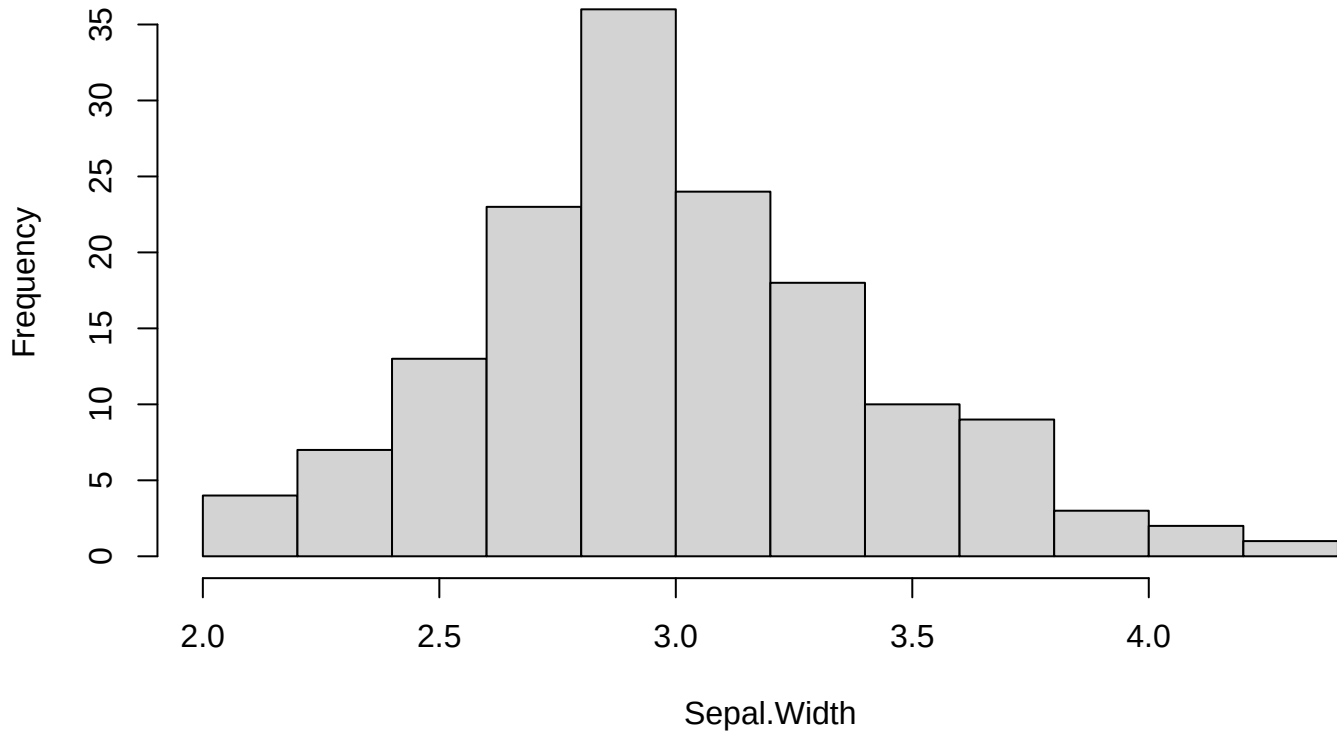
```
[1] 5.58 6.11 5.87 6.03 6.31 5.54 6.15 5.80
```

```
> barplot(HeavyWeight_clean, xlab='Flowers', ylab='Weight', main='Boxplot of  
Flowers Heavier than 5.5', col = hcl.colors(8, "Viridis"))
```

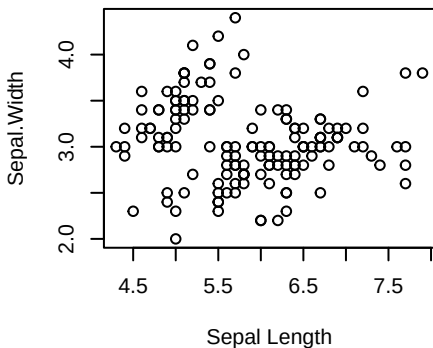
Hit <Return> to see next plot:

```
>
```

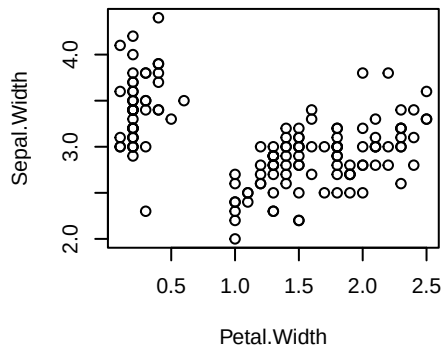
Histogram of Sepal.Width



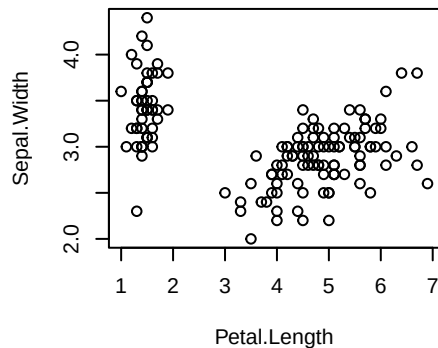
Sepal Width Vs Sepal Length



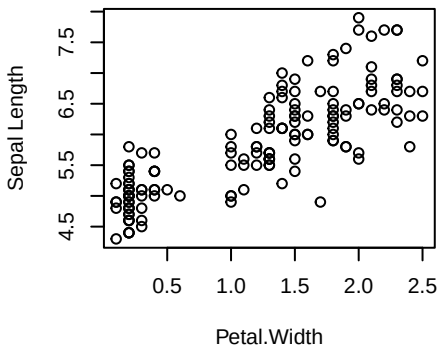
Sepal Width Vs Petal.Width



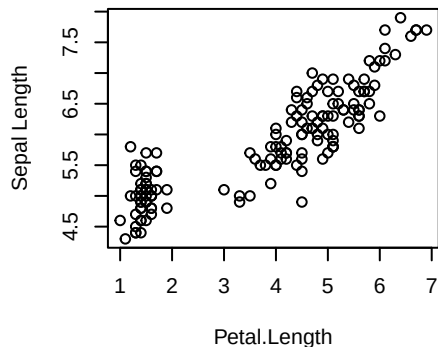
Sepal Width Vs Petal.Length



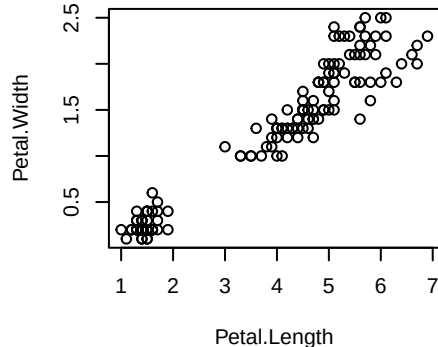
Sepal Length Vs Petal.Width



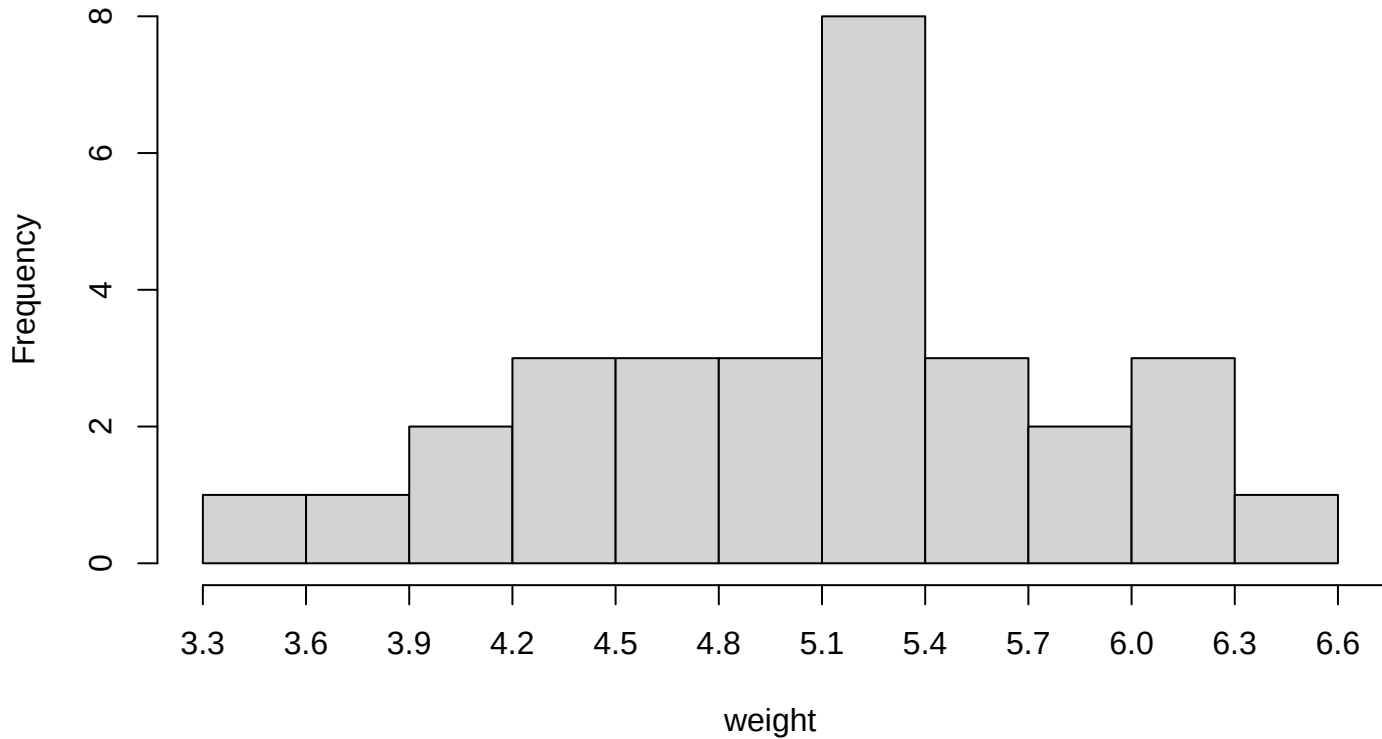
Sepal Length Vs Petal.Length

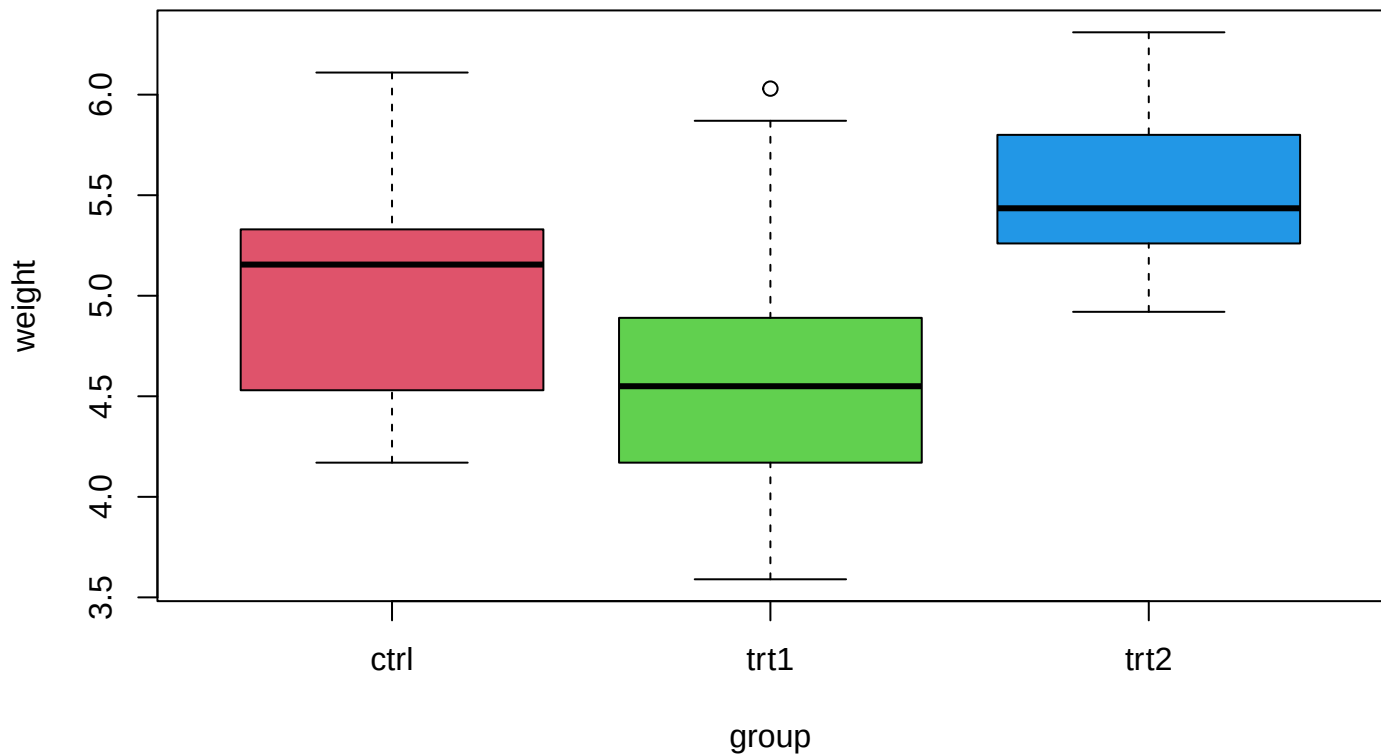


Petal.Width Vs Petal.Length



Plant Growth Weight





Boxplot of Flowers Heavier than 5.5

